NATIONAL CHILDREN'S SCIENCE CONGRESS (NCSC) 2026–27

FOUR CHART-PAPER HANDWRITING BLUEPRINTS

Document 05: Step-by-Step Handwriting Guides & Oral Speaking Notes for the 4 Standard NCSC Posters

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Important Instructions for the Child Scientist:
*NCSC presentation rules require **exactly four display charts** on standard chart paper (approx. 55 cm × 70 cm). These blueprints are NOT meant to be pasted as printed computer sheets. Instead, use these guides to draw and handwrite each poster neatly using colored sketch pens, bold markers, and a ruler. Keep paragraphs short and bulleted so judges can read them comfortably from a distance of 1.5 to 2.0 meters.*

POSTER 1: THE PROBLEM AND THE IDEA

Chart Sheet Size:	Standard Chart Paper (55 cm × 70 cm), White background
Suggested Color Palette:	Header: Dark Navy Blue / Dark Green | Subheadings: Forest Green | Text: Black
Space Allocation:	Top Banner: 15% | Left Column (Context & Problem): 42% | Right Column (Objectives & Principle): 43%

Exact Text & Structure to Handwrite on Poster 1:

1. MAIN HEADER (Large Bold Letters, Top Center):

KRISHI-DRISHTI: Offline-First Edge-AI Plant Health Screening Wand

Focal Theme: Science & Innovation for Sustainability | Sub-Theme: Food, Agriculture & Health

2. LOCAL AGRICULTURAL PROBLEM (Left Column, Top):

• Potato and tomato are major rabi cash crops in Bihar (Vaishali, Samastipur, Begusarai belts).
• Over 35% of crops are destroyed annually by Late Blight (*Phytophthora infestans*) and Early Blight (*Alternaria solani*).
• Smallholder farmers (< 2 ha) suffer complete crop loss within 48–72 hours of persistent winter fog and humidity.

3. WHY EXISTING SOLUTIONS FAIL (Left Column, Middle):

• *No Internet in Fields:* Cloud AI apps fail due to weak 4G/5G signals in remote rural tracts.
• *High Lab Cost:* Government PCR/ELISA tests cost ■15,000–■30,000 and take 7–14 days.
• *Panic Spraying:* Farmers spray expensive chemical cocktails blindly, wasting money and poisoning soil.

4. RESEARCH QUESTION & AIM (Right Column, Top):

• **Research Question:** Can an ultra-low-cost (< ■1,500), 100% offline microcontroller wand with microclimate telemetry provide preliminary foliar health screening without internet?
• **Aim:** Build a portable screening tool to help small farmers detect early blights and practice targeted sustainable interventions.

5. CORE SCIENTIFIC PRINCIPLE: THE PLANT DISEASE TRIANGLE (Right Column, Bottom):

[Hand-Draw a Large Triangle in Forest Green]:

Top Vertex: Susceptible Host (Tomato / Potato Plant)
Bottom-Left Vertex: Virulent Pathogen (*Phytophthora / Alternaria*)
Bottom-Right Vertex: Favorable Environment (Canopy Humidity > 85%, 18–24.5°C)
Center of Triangle: *DISEASE OUTBREAK OCCURS ONLY WHEN ALL 3 OVERLAP!*

• **Key Biological Distinction:** Late Blight is an OOMYCETE (water mold), not a true fungus! Chitin-inhibiting fungicides fail against it.

6. SUSTAINABILITY RELEVANCE (Bottom Banner, Full Width):

• Prevents toxic pesticide over-application | Protects groundwater and pollinating bees | Saves smallholder farmers ■2,000–■5,000 per acre.

Speaking Notes for Poster 1 (2-Minute Judge Walkthrough):

> *"Respected judges, in rural Bihar, potato and tomato crops face devastating destruction from foliar blights every winter. Small farmers cannot afford ■15,000 lab tests, and commercial smartphone apps fail because there is zero internet in rural fields. Out of panic, farmers spray toxic chemical fungicides blindly. Based on the Plant Disease Triangle, I asked: Can we build a portable, sub-■1,500 offline screening wand that combines camera imaging with canopy humidity sensing to guide farmers before they ruin their fields? This project directly addresses food security and agricultural sustainability."*

POSTER 2: HOW KRISHI-DRISHTI WORKS

Chart Sheet Size:	Standard Chart Paper (55 cm × 70 cm), White background
Suggested Color Palette:	Header: Slate Blue | Hardware Boxes: Light Gray with Dark Navy borders | Flowchart: Teal
Space Allocation:	Top Banner: 12% | Middle (Three Hardware Blocks): 45% | Bottom (Offline Flowchart & 4 OLED Screens): 43%

Exact Text & Structure to Handwrite on Poster 2:

1. MAIN HEADER (Large Bold Letters, Top Center):
SYSTEM ARCHITECTURE & OFFLINE EDGE OPERATION

2. THREE-TIER HARDWARE ARCHITECTURE (Hand-Draw 3 Clean Boxes Across Middle):
• **BOX 1: HANDHELD CONTROLLER WAND (ESP32 DevKit V1)**
- Non-blocking SCAN (GPIO 27) and NEXT (GPIO 26) buttons.
- 128×64 SSD1306 OLED screen (I2C: SDA 21, SCL 22).
- DHT11 canopy sensor (GPIO 18) monitoring temperature & humidity.
• **BOX 2: CAMERA UNIT WITH OPTICAL SHROUD (AI-Thinker ESP32-CAM)**
- OV5640 5MP camera sensor capturing UXGA (1600×1200) JPEGs.
- 3D-printed 8–10 cm cylindrical shroud enforcing fixed focal distance.
- High-intensity diffuse LED eliminating harsh shadows.
- Broadcasts offline Wi-Fi SoftAP mesh (192.168.4.1); writes to SD card.
• **BOX 3: POCKET EDGE-AI ENGINE (Farmer's Android Phone)**
- Connects to wand's local Wi-Fi (zero cellular data / zero internet).
- Runs quantized MobileNetV3 TFLite model (2.03 MB) in 4.9 ms.
- Dual-Stage Shannon Entropy Gating (H > 1.45 nats) rejects non-leaf frames.

3. STEP-BY-STEP OFFLINE WORKFLOW (Hand-Draw a Flowchart Across Bottom Left):
[Press SCAN] ■■> [DHT11 Reads Temp/RH] ■■> [ESP32-CAM Captures 1600x1200]
■■> [Phone AI Normalizes to 128x128] ■■> [MobileNetV3 Predicts in 4.9ms]
■■> [Entropy Check: Clear vs Uncertain] ■■> [OLED Wand Displays Carousel]

4. 4-PAGE OLED DISPLAY CAROUSEL (Hand-Draw 4 Small Screen Rectangles on Bottom Right):
• *Page 1 (Diagnosis):* 'TOMATO: LATE BLIGHT | Conf: 99.5% | SEVERE'
• *Page 2 (Microclimate):* 'Temp: 21.4°C | RH: 88.5% | HIGH SPORULATION RISK'
• *Page 3 (Organic Cure):* 'Spray Trichoderma viride @ 5g/L; remove infected foliage'
• *Page 4 (Chemical Spray):* 'Cymoxanil 8% + Mancozeb 64% @ 2.5g/L (Targeted intervention)'

Speaking Notes for Poster 2 (2-Minute Judge Walkthrough):

"Here is how KRISHI-DRISHTI works in the farmer's hands. We solved the hardware limitations of microcontrollers by dividing tasks: The lightweight ESP32 wand handles buttons, sensing, and the OLED display. The ESP32-CAM captures high-resolution leaf photos through an 8–10 cm optical shroud that eliminates blur. To avoid draining the microcontroller, inference runs on the farmer's pocket smartphone via a local Wi-Fi mesh. The phone runs our 2.03 MB MobileNetV3 model in less than 5 milliseconds without internet, and immediately returns a 4-page diagnosis and organic cure directly to the wand screen."

POSTER 3: EXPERIMENTS AND RESULTS

Chart Sheet Size:	Standard Chart Paper (55 cm × 70 cm), White background
Suggested Color Palette:	Header: Charcoal / Dark Red | Tables: Navy borders | Graph Bars: Teal & Dark Blue
Space Allocation:	Top Banner: 12% | Left Column (Verified Tables & Zero-Leakage Audit): 48% | Right Column (Grouped Bar Chart & Pie Chart): 52%

Exact Text & Structure to Handwrite on Poster 3:

1. MAIN HEADER (Large Bold Letters, Top Center):
EXPERIMENTAL VALIDATION & BENCHMARK PERFORMANCE

2. ZERO-LEAKAGE DATASET QUALITY AUDIT (Left Column, Top):
• 7,710 records audited & reconciled into exactly **6,965 unique training hashes**.
• 5 duplicate pairs discovered in upstream Kaggle dataset—all verified to share identical labels (0 cross-class conflicts).
• 100% cryptographic holdout exclusion of all 60 benchmark specimens and 93 PlantDoc field images.

3. VERIFIED BENCHMARK RESULTS TABLE (Left Column, Bottom):
[Hand-Draw This Clean Table]:
• *Production Daemon:* 8 Canonical Leaves ■■> **100.0% (8/8)** | Latency: 4.9 ms
• *Botanical Benchmark:* 60 Leaves (10/class) ■■> **88.33% Strict 6-Class (53/60)** | **92.98% Accepted (53/57)**
• *Pathogen Family Match:* 54/60 (**90.0%**) — 1 cross-crop Late Blight leaf matched oomycete biology.
• *PlantDoc Outdoor Set:* 93 Unseen Field Photos ■■> **33.33% Raw (31/93)** ■■> **55.36% Accepted Diagnostic (31/56)**
• *Dual-Stage Gating:* Safely rejected 37 ambiguous outdoor frames (39.78% rejection rate).

4. HAND-DRAWABLE GROUPED BAR CHART (Right Column, Top):
[Draw a Clear Bar Chart with Y-axis: Accuracy (0 to 100%), X-axis: Test Benchmarks]:
• Benchmark 1 (60 Botanical Leaves): Bar 1 (Raw) = 88.33% | Bar 2 (Accepted) = 92.98%
• Benchmark 2 (93 PlantDoc Field Leaves): Bar 1 (Raw Baseline) = 33.33% | Bar 2 (Accepted Gated) = 55.36%
• *Key Insight:* Shannon entropy gating removes low-quality/blurry captures, increasing diagnostic precision by +22.03%!

5. MATHEMATICALLY VERIFIED PLANTDOC TRIAGE PIE CHART (Right Column, Bottom):
[Exact Mathematical Calculations for Drawing with a School Protractor]:
Formula: $\text{Angle} = (\text{Count} / 93) \times 360^\circ$
• **Sector 1 (Accepted Correct):** 31 / 93 = 33.33% ■■> Exact Angle: **120.0°** (Draw with protractor: **120°**) [Color: Emerald Green]
• **Sector 2 (Gated Uncertain):** 37 / 93 = 39.78% ■■> Exact Angle: **143.2°** (Draw with protractor: **143°**) [Color: Amber Yellow]
• **Sector 3 (Accepted Error):** 25 / 93 = 26.88% ■■> Exact Angle: **96.8°** (Draw with protractor: **97°**) [Color: Muted Red]
• **Angle Sum Check:** $120^\circ + 143^\circ + 97^\circ = 360^\circ$ *(Adds up to exactly 360 degrees!)*
Protractor Tip: Draw a circle (radius 6 cm). Mark horizontal baseline. Measure 120° for Sector 1, then from that line measure 143° for Sector 2. The remaining slice is exactly 97°!

Speaking Notes for Poster 3 (2-Minute Judge Walkthrough):

"Judges often ask if student AI models are truly tested or just memorized. On this poster, I show our cryptographic proof. We audited 7,710 records to guarantee zero training overlap. On 60 clean botanical specimens, our model achieved 88.33% strict accuracy and 92.98% on accepted frames. More importantly, on 93 tough outdoor field photos from PlantDoc, raw accuracy was only 33.33%. But our Shannon entropy filter safely caught and rejected 37 ambiguous frames, boosting accuracy on accepted frames to 55.36%. In medicine and agriculture, knowing when you are uncertain is just as vital as being right!"

POSTER 4: FINDINGS, LIMITATIONS AND FUTURE SCOPE

Chart Sheet Size:	Standard Chart Paper (55 cm × 70 cm), White background
Suggested Color Palette:	Header: Dark Green / Navy | Success Callouts: Forest Green | Limitations: Warm Amber
Space Allocation:	Top Banner: 12% | Left Column (Successes & Failure Cases): 48% | Right Column (Limitations & Future Scope): 52%

Exact Text & Structure to Handwrite on Poster 4:

1. MAIN HEADER (Large Bold Letters, Top Center):
REAL-WORLD FINDINGS, LIMITATIONS & SUSTAINABILITY IMPACT

2. WHAT WORKED WELL (Left Column, Top):
• *100% Offline Mesh:* Complete telemetry and diagnosis transfer with zero internet and zero cloud fees.
• *Blazing Latency:* 4.9 ms inference on budget hardware; total scan-to-OLED round trip < 2.5 seconds.
• *Shroud Stabilization:* Enforced the 8–10 cm lens sweet spot, eliminating blurry macro captures in the field.
• *Microclimate Spray Suppression:* Successfully suppressed high-cost chemical sprays when humidity was < 85%.

3. HONEST FAILURE CASES DISCOVERED (Left Column, Bottom):
• *Failure Case 1 (Harsh Shadows):* Nocturnal unshielded torch flash created steep shadows on healthy tomato leaves that the model misidentified as Late Blight.
• *Failure Case 2 (Alien Foliage):* An out-of-distribution tea leaf was accepted as Late Blight with 87.0% confidence! The shroud fixed distance, but could not reject alien green foliage.

4. CRITICAL SCIENTIFIC PRINCIPLES (Right Column, Top):
• **Pathogen Match ≠ Exact Classification:** Cross-crop late blight predictions share identical *Phytophthora infestans* biology, but are reported separately from exact 6-class passes.
• **Shroud is an Acquisition Constraint, Not an OOD Shield:** Enforces optical geometry, but cannot guarantee non-crop plant exclusion.
• **Microclimate Advisory ≠ Proof of Absence:** Instantaneous RH < 85% is a risk advisory based on Wallin/Hyre models, not biological proof that spores are absent.

5. SUSTAINABILITY & FARMER IMPACT (Right Column, Middle):
• *Economics:* < ■1,500 device pays for itself by preventing just 1 unneeded fungicide spray (saving ■1,500–■2,500).
• *Ecological Health:* Promotes organic Trichoderma treatments; reduces toxic runoff into rural Bihar groundwater.

6. THE NEXT DEFINING MILESTONE (Right Column, Bottom):
• **Authentic Hardware Field Captures:** Transitioning from internet photos to hundreds of images captured directly by the KRISHI-DRISHTI wand in rural Bihar fields.
• Future additions: Open-set recognition (OSR), continuous leaf wetness logging, and voice advice in Bhojpuri and Maithili.

Speaking Notes for Poster 4 & Closing Defense (2-Minute Judge Walkthrough):

"To conclude, KRISHI-DRISHTI is not a theoretical AI claim; it is a physical engineering tool tested under real constraints. We proved that an ESP32 wand and a low-cost smartphone can perform offline foliar screening in under 5 milliseconds for under ■1,500. We do not claim 100% accuracy. We openly showed where night shadows caused false alarms and how an alien tea leaf fooled our closed-set model. That is why our next major milestone is large-scale field validation directly with local Bihar farmers using the physical wand. By empowering small farmers with early organic guidance, KRISHI-DRISHTI turns science into sustainable agricultural impact. Thank you!"